Generative AI for Synthetic Industrial Surface Defect Generation Under Low-Data Constraints

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The goal of this continuation project is to extend my capstone work on generative AI for synthetic industrial surface defect generation under low-data constraints. The broader motivation is that in industrial inspection settings, real defect images are often scarce, imbalanced, expensive to document, or difficult to share due to proprietary manufacturing constraints. This limits the ability to train robust computer vision models for defect detection and inspection tasks.

During the first phase of the capstone, I implemented and evaluated conditional DCGAN-based models for generating synthetic grayscale metal surface images across three classes: normal, scratch, and spot. The experiments showed that the model could learn shared metal surface texture and some class-specific cues, but it struggled with generating diverse and structurally realistic defects, especially scratches. Spot-like defects were easier to learn than scratch-like defects, while diversity remained the main limitation. These findings motivate the continuation of the project toward more controllable and realistic defect-generation approaches.

Project Goals

1. **Controlled cGAN Proof-of-Concept:** Implement a simplified conditional GAN experiment using hand-drawn scratch-like images on plain or minimally textured backgrounds. This will test whether the cGAN architecture can learn a diverse range of such scratch-like spatial patterns when the complexity of real metal texture is removed.

2. **Image-to-Image Translation for Defect Generation:** Explore image-to-image translation (such as CycleGAN) approaches for converting normal surface images into defect-containing images. This direction is motivated by the limitation that generating every pixel from scratch using only a class label is difficult under low-data conditions.

3. **Defect-Aware or Segmentation-Guided Generation:** Investigate simple defect-aware conditioning strategies, such as approximate defect masks or spatial guidance, to improve control over where defects appear and what structure they take.

4. **Improved Evaluation and Checkpoint Selection:** Develop a more structured evaluation framework for generated outputs using visual realism, class consistency, and diversity. If feasible, classifier-assisted evaluation will also be explored to help identify useful generated outputs and intermediate checkpoints.

Work Completed So Far

In the initial capstone phase, I built a conditional DCGAN pipeline in TensorFlow/Keras for synthetic surface defect image generation. The pipeline included image resizing, grayscale conversion, normalization, label encoding, class balancing, generated sample logging, checkpointing, and loss tracking.

The first experiment used one three-class cDCGAN for normal, scratch, and spot images. This baseline showed that the model learned general metal surface texture more easily than defect-specific structure. To better understand this limitation, I simplified the task into binary experiments: normal versus scratch and normal versus spot.

The binary experiments showed that class separation improved compared with the three-class baseline, especially after adding defect-specific enhancement, stratified batching, and mild augmentation. However, the outputs still lacked diversity, and scratch-like structures remained difficult to model. These results suggest that the current label-conditioned cDCGAN provides a useful baseline, but stronger conditioning and alternative generative approaches are needed for more controllable defect synthesis.

Proposed Work for the Continuation Semester

The continuation semester will focus on extending the project beyond the current cDCGAN baseline. The first planned task is a simplified proof-of-concept experiment using hand-drawn scratch-like images. This experiment will isolate the challenge of scratch generation from the complexity of real metal texture and help determine whether the current architecture can learn scratch-like spatial patterns under controlled conditions.

After this proof-of-concept, I will move toward image-to-image translation approaches. Instead of generating the entire metal surface image from latent noise, these methods can start from a normal surface image and translate it into a defect-containing image. This may be more appropriate for industrial defect synthesis because the background surface texture can be preserved while the model focuses on adding realistic defects.

I will also explore defect-aware or segmentation-guided strategies inspired by defect-transfer methods. The goal is to test whether adding spatial information, such as approximate masks or defect-location guidance, improves control over defect placement, size, and structure. This is especially relevant for scratch defects, where a simple class label does not provide enough information about orientation, length, or location.

Finally, I will improve the evaluation process. In the first phase, I observed that the best-looking generated outputs did not always occur at the final epoch, and GAN loss curves alone were not sufficient to judge output quality. During the continuation, I will use a more structured evaluation framework based on visual realism, class consistency, and diversity. If feasible, I will also train or use a classifier-assisted evaluation method to help compare generated outputs across checkpoints and experiments.

Expected Deliverables

By the end of the continuation semester, I plan to complete the following deliverables:

- A simplified cGAN proof-of-concept experiment using hand-drawn scratch-like images.
- One or more image-to-image translation experiments for normal-to-defect generation.
- Initial defect-aware or segmentation-guided experiments for improved controllability.
- A structured evaluation framework comparing visual realism, class consistency, and diversity.
- Updated generated image results and visual comparisons across model families.

- A cleaned and organized codebase with reproducible scripts, experiment folders, and output summaries.
- A final capstone report documenting methodology, experiments, results, limitations, and future work.

Expected Contribution

My contribution during the continuation semester will be to design and implement the additional experiments, prepare the datasets and preprocessing pipelines, train and evaluate the models, compare results across approaches, and document the findings in a final report.

The expected outcome is not necessarily a production-ready synthetic defect generator, but a stronger experimental framework for understanding the trade-offs between class-conditioned generation, image-to-image translation, and spatially guided defect generation under low-data industrial settings. This work may also provide the foundation for a future conference-style submission or technical article on synthetic defect generation for industrial computer vision.